

Tactile-Enabled Contact Insertion for Automated Wire-Harness Assembly

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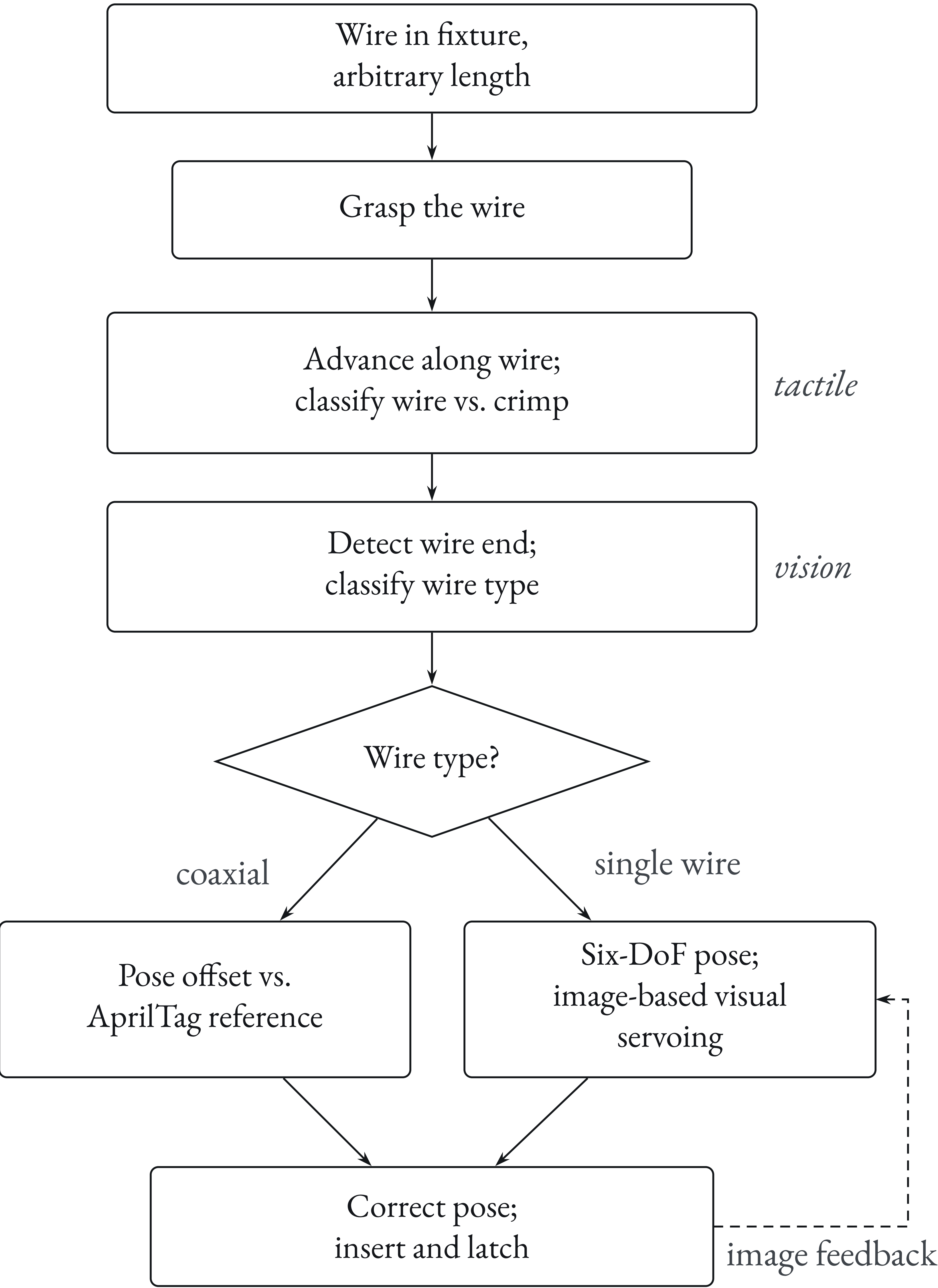
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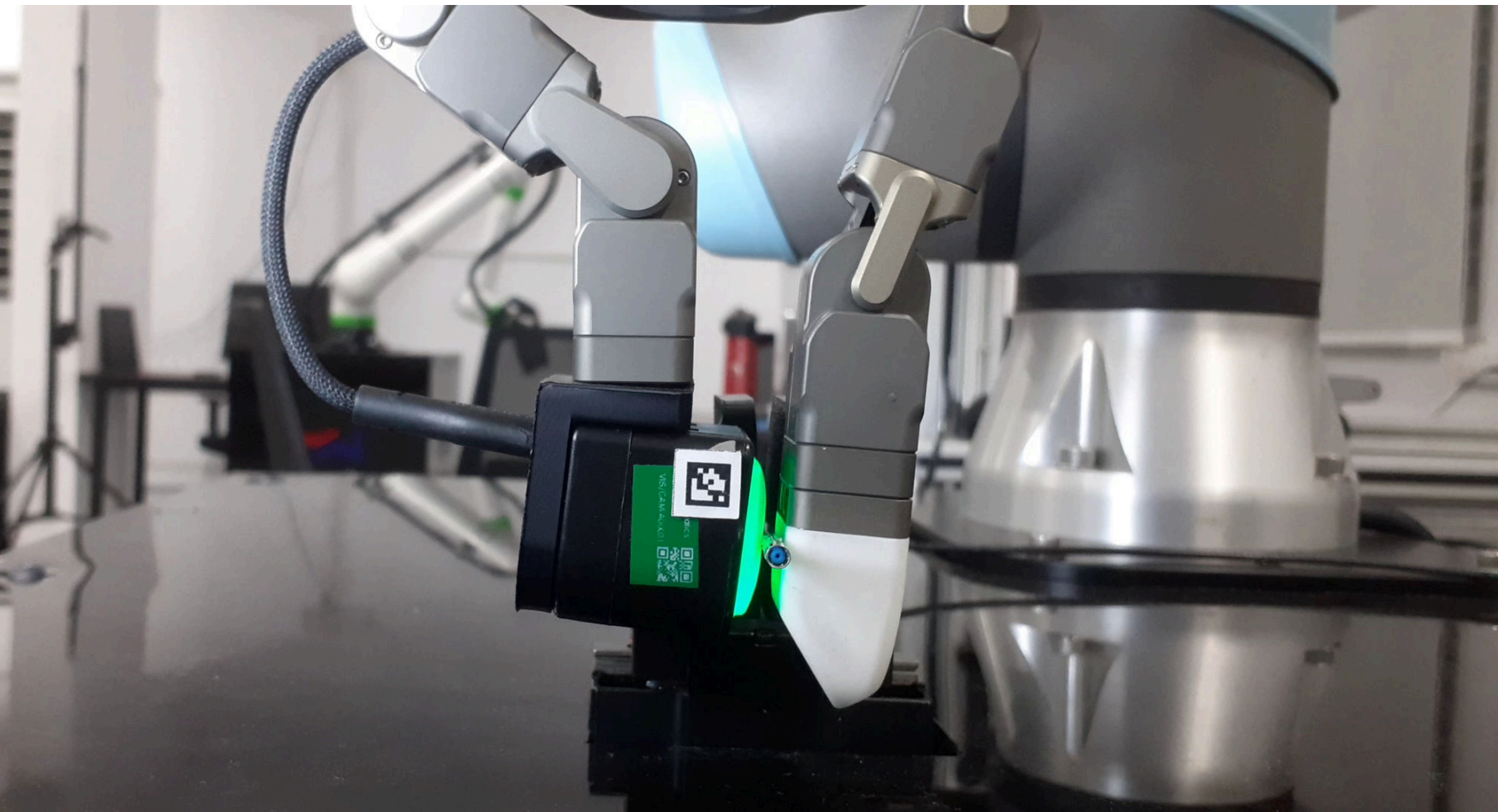
5,000	3,500 m	60 kg	4 min 30 s
individual parts in a harness	of installed cable	per vehicle	to build one module by hand

Harness production is largely manual, and this step is contact-rich and sub-millimetre. Robotic manipulation leans on *vision and open-loop strategies*; here one finger of the gripper is a DIGIT visuo-tactile sensor. The task is an externally judged industrial benchmark. The focus is on the insertion step.

The insertion sequence



The system



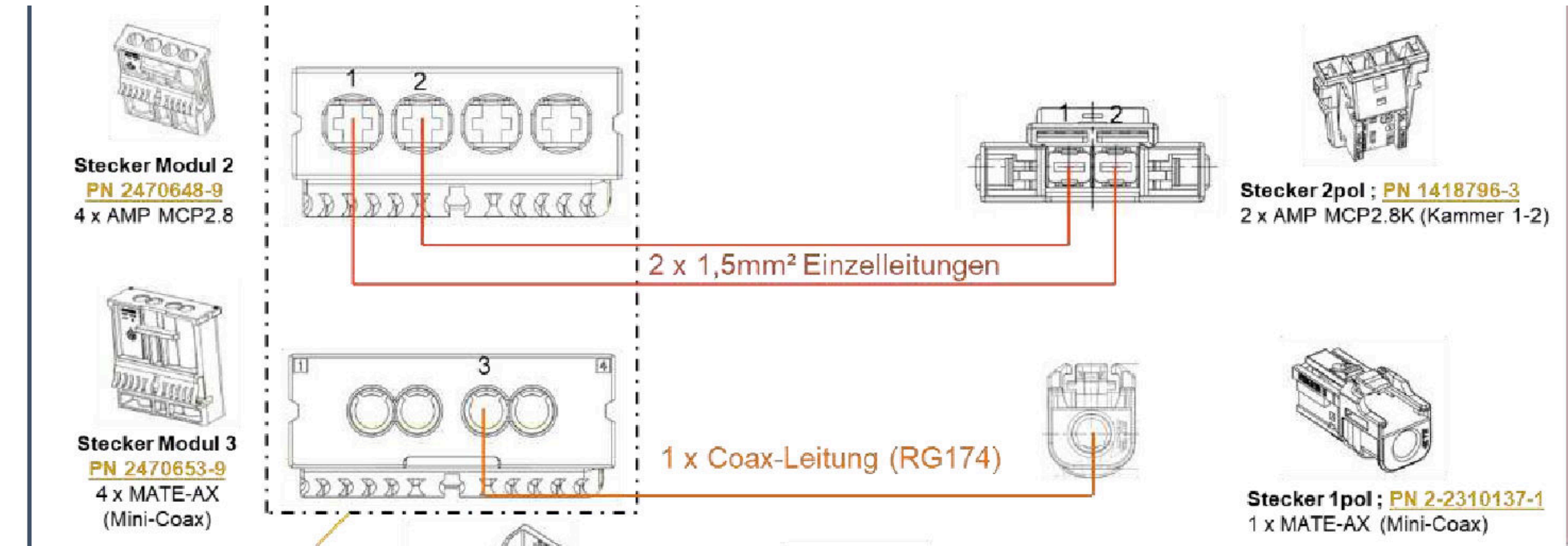
Gripper with tactile sensor

A Tesollo DG-2F gripper on a Universal Robots e-series arm, one finger replaced by a Meta DIGIT. There is no force–torque sensor and no force feedback.

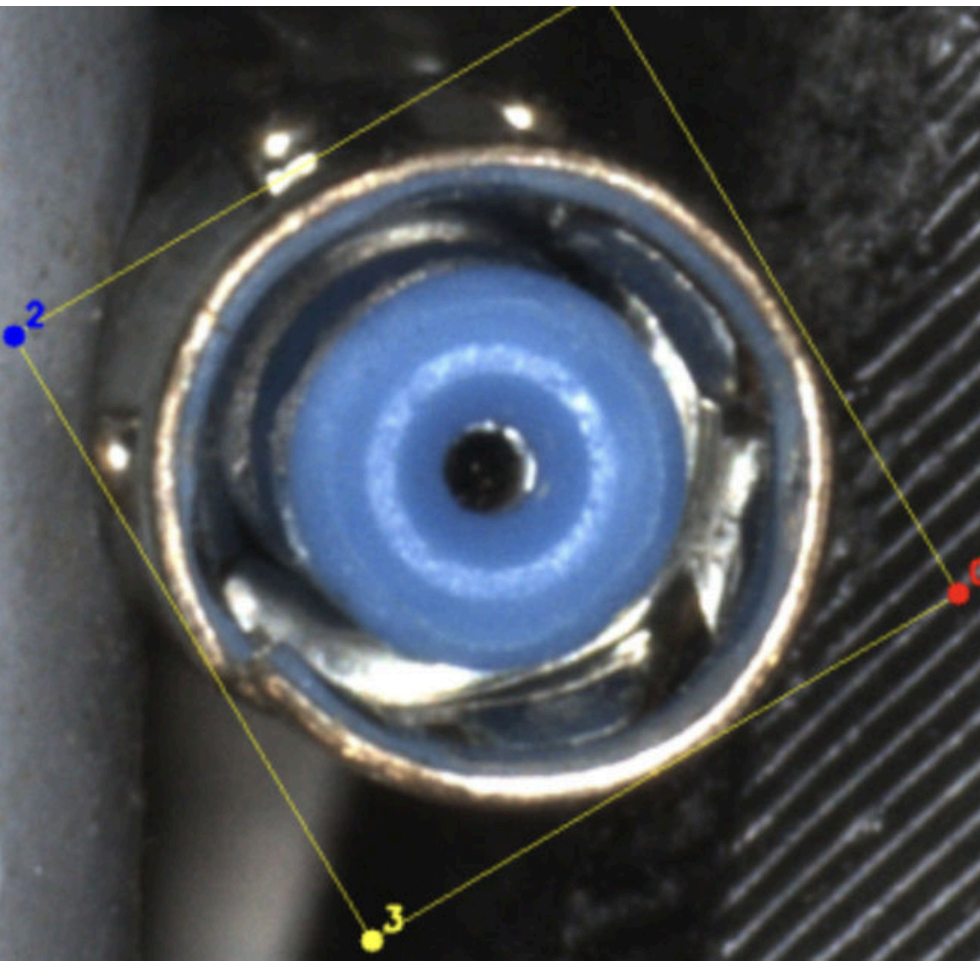
Once the fingers close, the camera can no longer see where the grasp sits. The DIGIT runs on its green channel alone; a classifier decides at the contact whether the fingers hold bare wire or a crimped terminal. Classical control performs the insertion itself.

60 / 60

trials, with no insertion failure attributed to the tactile stage.



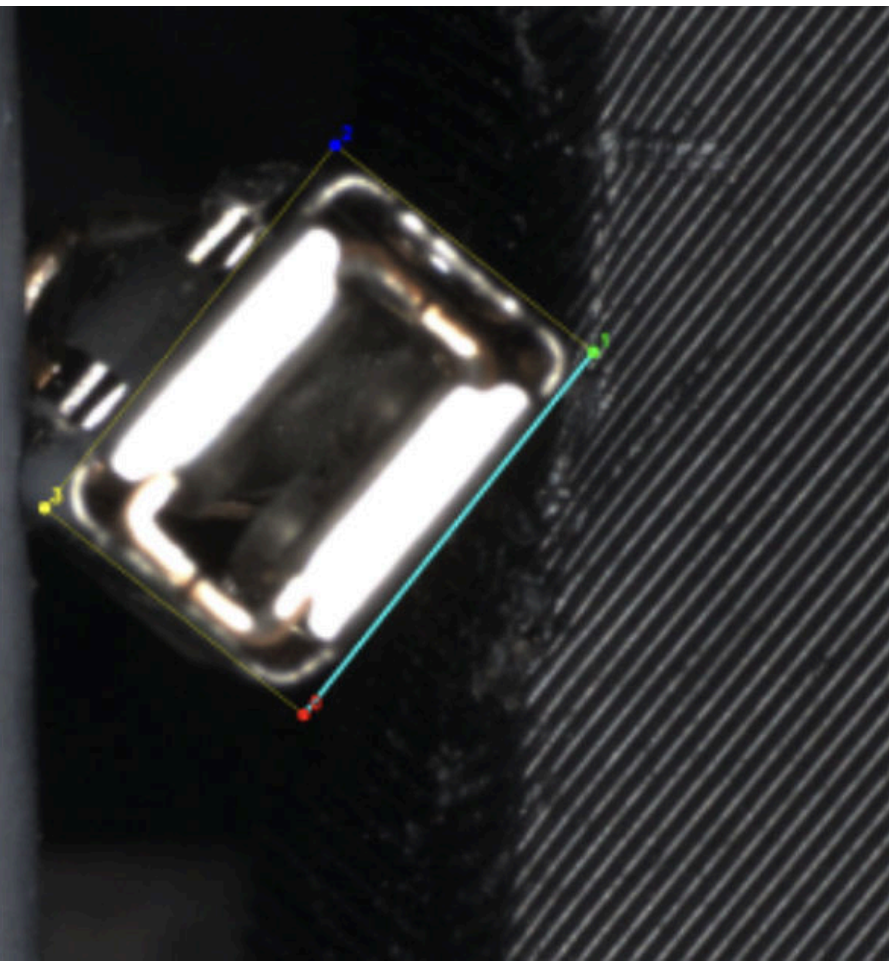
Contact geometry



Coaxial contact

29/30

insertions succeeded



Rectangular contact

15/30

insertions succeeded

The detected class selects the correction. A circular face needs only a planar offset against an AprilTag reference. A rectangular face must have its rotation resolved, so the loop closes on image features, under image-based visual servoing.

Failures were translational: the contact jammed at the cavity mouth, or passed just above or below it.

The tactile decision is only as good as the grasp beneath it.

The harness module

Modul 2 takes the single wires, Modul 3 the coaxial line. Each contact has its own numbered cavity.